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ABSTRACT

The paleomagnetism of middle to late middle Miocene volcanic rocks from regions of the western Transverse Ranges suggests that large amounts of subsequent clockwise tectonic rotation has occurred. We studied rocks from five sites in the western Santa Monica Mountains, Conejo Hills, and Anacapa Island. East declinations of from 64° to 81° were found, compared to expected Miocene declinations of 0°, and inclinations were too shallow by up to 21° for the present latitude. Preliminary data from Santa Cruz and San Miguel Islands suggest similar but larger rotations. Stratigraphically higher units show the least amount of rotation, which may indicate that about 60° of rotation occurred primarily during middle and late Miocene time. North-pointing directions can be restored by rotating the northern Channel Islands and Santa Monica Mountains counterclockwise about a pivot near the eastern end of the range. This restoration also changes paleocurrent directions in Eocene and Oligocene rocks from north to west, which is consistent with data from surrounding regions. Shallow remanent inclinations suggest northward movement of this block of ~10° since early Miocene time.

INTRODUCTION

The history of the Transverse Range Province of southern California has long been an enigma to earth scientists because the east-west trend of mountains and basins lies across the predominant northwest-southeast trend of the Coast Ranges and Sierra Nevada. Plate-tectonics applications account for the Transverse Ranges as resulting from Pliocene-Pleistocene compression associated with the opening of the Gulf of California (Crowell, 1975; Anderson, 1971). Hadley and Kanamori (1977) discovered that the Transverse Ranges lack a crustal root and are underlain by an anomalous high seismic velocity structure (ridge?) in the mantle. This suggests that the Transverse Ranges formed in a weak thin zone of the lithosphere, probably beginning in Pliocene time. However, older east-west–trending features within the province cannot be explained by the Pliocene-Pleistocene compression. Jones and Irwin (1975), in comparing belts of Mesozoic rocks from Oregon to Baja California, found east-west–trending foliations, fold axes, and major lithic belts of slate and andesite within the Transverse Ranges, whereas elsewhere correlative belts have a north-south orientation. They thus proposed a major rotation (which could presumably be in either direction) in post-Kimmeridgian time. Baird and others (1974) also discuss these east-west–trending features but concluded that the amount of rotation necessary was improbable and that the structure had formed with an east-west trend. Jones and others (1976) also suggested clockwise rotation of the Santa Monica slate and Santa Cruz Island in Late Cretaceous or Cenozoic time, but they did not specify the exact amount. Hamilton (1978) proposed that a block including the western Transverse Ranges and at least Anacapa and Santa Cruz Islands rotated clockwise about an eastern pivot from the peninsula ranges.

Large tectonic rotations are being increasingly suggested by paleomagnetic data from western North America (Beck, 1976). In general, the divergent paleomagnetic directions are deflected clockwise and show a shallowing of inclination relative to expected field directions for stable North America. Simpson and Cox (1977) reported divergent paleomagnetic directions for Eocene volcanic rocks in the Oregon-Washington coast range, of 50° to 70° east of the expected direction. Greenhaus and Cox (1978) noted clockwise deflections for the Miocene intrusives of the Morro Rock-Islay complex. Beck (1976) proposed that these divergent directions are due to regional shear between North America and oceanic plates as discussed by Atwater (1970). Nearer to the present study, Teissere and Beck (1973) found the paleomagnetic pole for Cretaceous southern California batholith rotated 26° clockwise from its expected position.

Miocene and Oligocene volcanic rocks are common over large areas of the Transverse Ranges and are ideal for paleomagnetic study because of their magnetic stability and high magnetic intensity. The Conejo Volcanics (Taliaferro and others, 1924) occur in the Santa Monica Mountains, Conejo Hills (Shelton, 1954), and on Anacapa Island (Scholl, 1960). These rocks consist of a sequence of volcanic breccia, tuff breccia, pillow lavas, and massive flows, which are intruded by dikes, sills, and hypabyssal intrusives. Andesitic and basaltic rocks predominate, although dacitic to rhyolitic compositions occur, generally toward the top of the section. Submarine to subaerial origin for the volcanics suggests an emerging submarine volcanic structure in the Conejo Hills area (Williams, 1977). Potassium-argon dates on these volcanic rocks range from 16.1 to 13.1 m.y. (Turner, 1970).

PALEOMAGNETIC RESULTS

More than 200 samples were obtained by portable gasoline-powered drill from five widely spaced locations in the Santa Monica Mountains, Conejo Hills, and east Anacapa Island, en-

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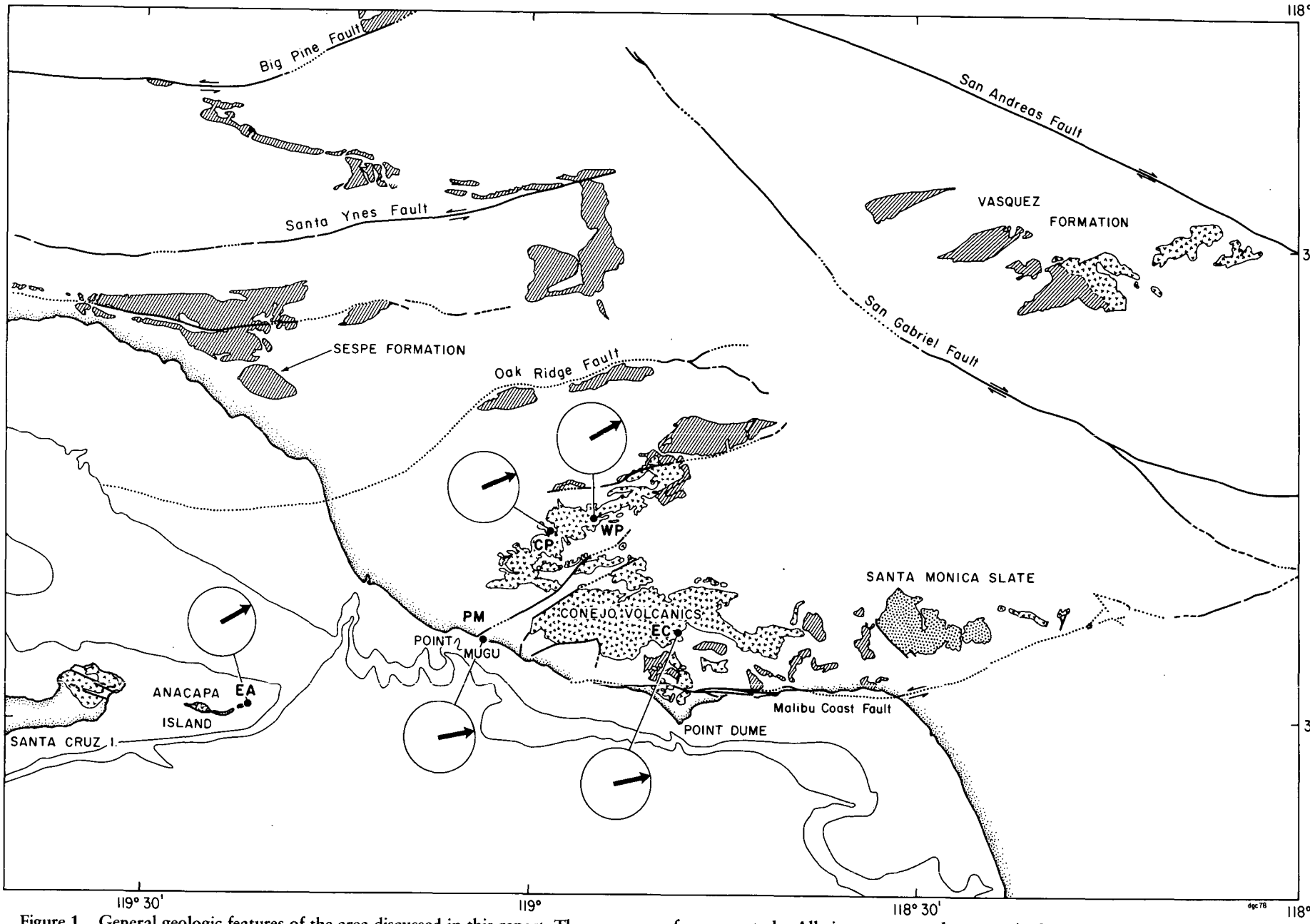


Figure 1. General geologic features of the area discussed in this report. The arrows in the open circles show the paleomagnetic declinations in Miocene rocks determined from our study. All sites are reversely magnetized except PM (Point Mugu). EA (East Anacapa), EC (Encinal Canyon, CP (Camarillo Park), and WP (Wildwood Park).

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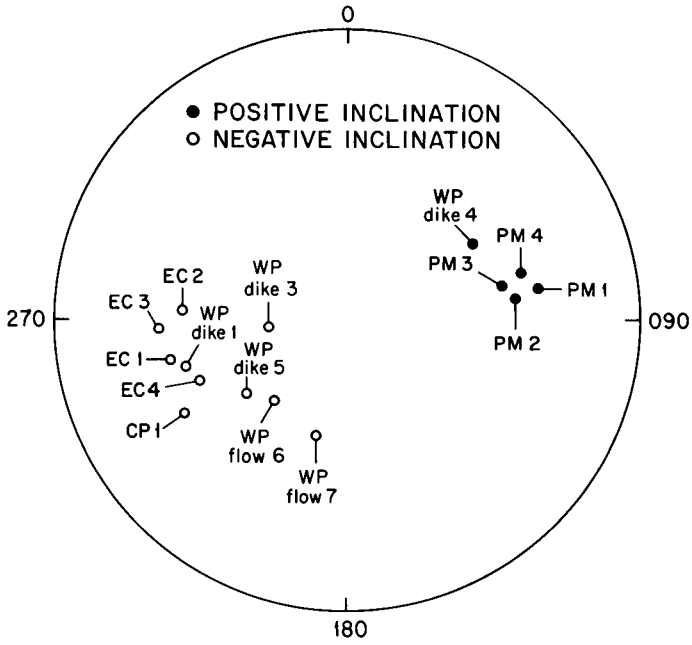


Figure 2. Stereoplot remanent magnetization directions determined for rock units in the Santa Monica Mountains.

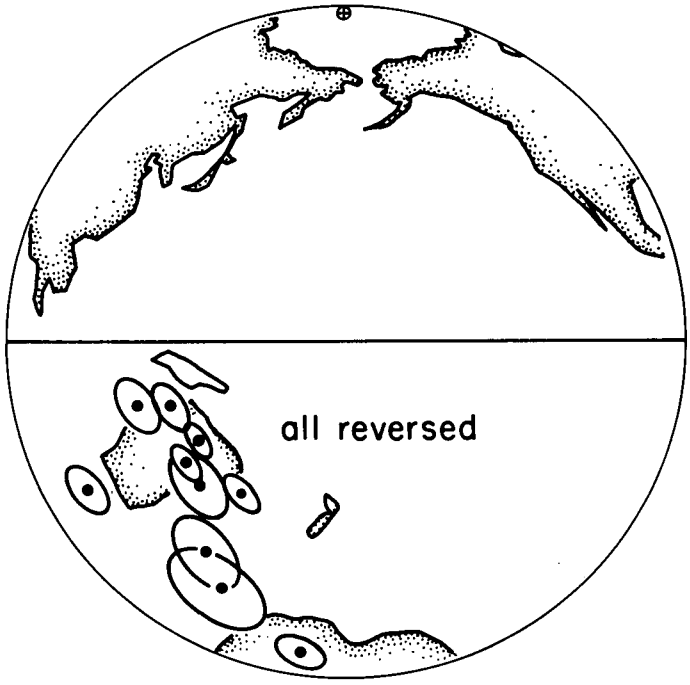


Figure 3. South virtual geomagnetic poles (VGP) of reversely magnetized rock units from Camarillo Park, Encinal Canyon, and Wildwood Park, Santa Monica Mountains. The ovals are 95% confidence limits.

compassing an area of 400 km². Each sample was individually step-wise demagnetized with a Schonstedt alternating field demagnetizer. Peak fields necessary for cleaning varied from 100 to 600 Oersteds, although some samples required higher fields for optimum cleaning. This procedure was found to be desirable, as optimum peak cleaning fields varied from unit to unit and even within a single dike or flow, depending on position of sample with respect to the unit margin (Petersen, 1976). Cleaned directions were

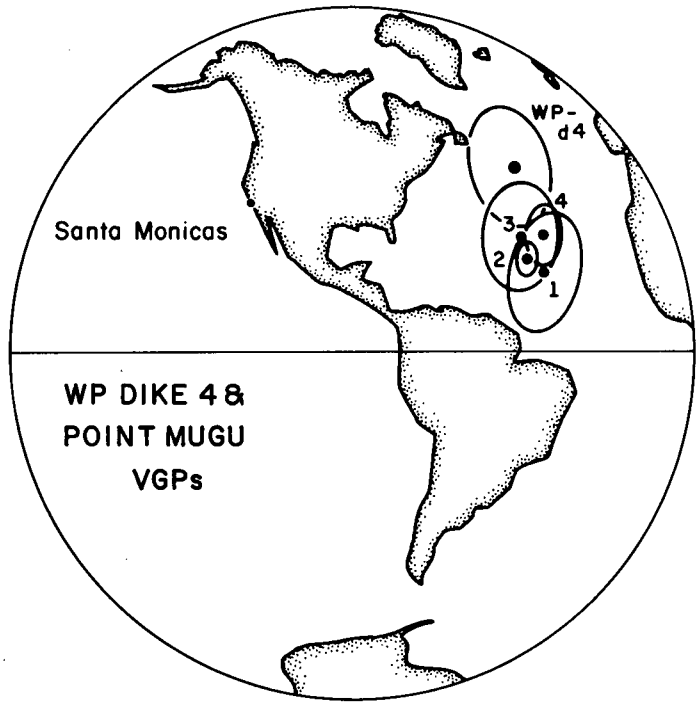


Figure 4. North virtual geomagnetic poles for normally magnetized rocks from Point Mugu and Wildwood Park, Santa Monica Mountains.

selected with the aid of stereographic plots and stability index calculations of Symons and Stupevesky (1974). Our Schonstedt spinner magnetometer was calibrated by standards from the U.S. Geological Survey. Also, paleomagnetic directions of two samples measured at the paleomagnetic laboratory at Stanford University and on our magnetometer were identical within a few tenths of a degree.

Thirteen dikes and two flows from the four locations in the Santa Monica Mountains show deflection of declination to the east from the expected paleomagnetic late Tertiary pole of McElhinny (1973) (Figs. 1 and 2). These deflections vary from 14° to 82° east in declination. The present dipole inclination for this area is about 53.5°. Inclinations for the units measured vary from 34.6° to 68°, but the average is near 45°. Figure 2, and the VGP (virtual geomagnetic pole) data (Figs. 3 and 4; Table 1) show an antipodial distribution demonstrating that the reversal test is met.

Dike 2 and flows 1 through 5 at Wildwood Park (WP) were not used in calculating mean directions. Although stable directions were found (Table 1) after alternating field demagneticzation, thermal demagnetization, hysteresis measurement, and polished section studies showed that these units have a secondary chemical remanence. The remaining two flows at WP, 6 and 7, are more consistent with the other data except that they show progressively less deflection of declination and the inclinations are close to the expected inclination (Fig. 2; Table 1). These flows are stratigraphically the youngest units sampled in our study region.

The results from east Anacapa Island are from ten massive flows of middle Miocene age which encompass about 60 m of section. The strata have been tilted 20° to the north, about an east-west axis. All the samples from east Anacapa Island are reversely magnetized and show deflections of declination of 50° to 76° clockwise from the expected direction, and the inclinations are too shallow by about 20° (Fig. 5; Table 2).

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TABLE 1. PALEOMAGNETIC DATA FOR THE SANTA MONICA MOUNTAINS AND CONEJO HILLS

Unit	No.	Mean inclination	Mean declination	Kappa	Alpha 95	Virtual geomagnetic pole Lat	Long
PM							
Dike 1	5	34.7	82.3	94.3	7.9	16.7 N	41.1 W
Dike 2	6	43.5	82.1	457.4	3.1	20.0 N	46.7 W
Dike 3	5	46.4	77.3	60.6	9.9	25.0 N	46.7 W
Dike 4	4	39.9	74.2	238.3	6.0	25.1 N	40.3 W
EC							
Dike 1	6	−38.8	256.8	286.2	4.0	22.6 S	139.3 E
Dike 2	8	−43.7	273.1	64.5	7.0	11.6 S	128.1 E
Dike 3	7	−36.1	266.4	103.1	6.0	13.9 S	136.3 E
Dike 4	9	−45.0	247.1	93.1	5.4	32.7 S	139.2 E
CP							
Dike 1	22	−35.8	239.2	47.1	4.6	36.1 S	150.4 E
WP							
Dike 1	9	−42.6	253.8	225.0	3.4	26.4 S	138.0 E
Dike 3	9	−68.0	266.0	239.2	3.3	28.3 S	106.5 E
Dike 4	8	48.6	59.2	28.7	10.5	40.2 N	40.7 W
Dike 5	5	−54.8	232.6	233.7	5.0	47.3 S	134.1 E
Flow 6	5	−59.8	220.9	106.8	7.4	57.2 S	127.4 E
Flow 7	6	−56.1	194.1	358.3	3.5	78.3 S	134.7 E
Mean	13	44.8	75.4	45.9	6.2	25.9 N	44.4 W*
Mean	15	47.6	70.9	25.7	7.7	30.5 N	44.7 W†
			WP data not used in averages				
Dike 2	11	58.4	311.2	31.1	8.3	51.0 N	172.9 E
Flow 1	8	63.9	172.5	51.2	6.1	9.8 S	113.6 W
Flow 2	4	59.2	163.6	557.2	3.0	14.2 S	106.0 W
Flow 3	3	63.1	167.6	558.0	5.2	10.3 S	110.0 W
Flow 4	10	58.6	175.8	254.5	3.0	16.3 S	115.5 W
Flow 5	6	62.7	170.1	126.3	6.0	11.1 S	111.6 W

Note: data listed in order of decreasing age.
* Flows 6 and 7 not included; Northern Hemisphere equivalent.
† Flows 6 and 7 included; Northern Hemisphere equivalent.

DISCUSSION

Our results indicate that the Santa Monica and Anacapa region has been rotated clockwise by about 70° since middle Miocene time and possibly transported northward about 10° in latitude (Table 3). Because both normal and reversely magnetized units are observed (Tables 1 and 2), an excursion of the field is an unlikely explanation of the data. Tectonic tilting can also be ruled out, because the locations sampled are apparently structurally simple, and the corrected results are all consistent. In the areas sampled, field relationships indicate that the section was tilted after the main period of volcanism (Sonneman, 1956; Williams, 1978). Therefore, dikes were corrected for the attitude of the country rock.

The uppermost lava flows, 6 and 7, sampled at WP indicate only 40° and 14° rotation, and stratigraphically lower units show progressive increases in the amount of apparent rotation (Figs. 2 through 4; Tables 1 and 3). All of the rocks studied fall into the middle Miocene period; Point Mugu rocks are stratigraphically lowest. Dike 4 at WP has a K-Ar age of 10.1 ± 2 m.y. (D. Krummenacher, 1977, personal commun). This dike is older than flow 6 at WP. This may indicate that the rotation was taking place during the time the units were formed and ended shortly after the time of the uppermost flows, late middle Miocene (Williams, 1977). Also, Tables 1 and 3 suggest a systematic steepening of inclination with decreasing age, possibly due to latitude increases. These observations may also in part be due to secular variation. At each site, not

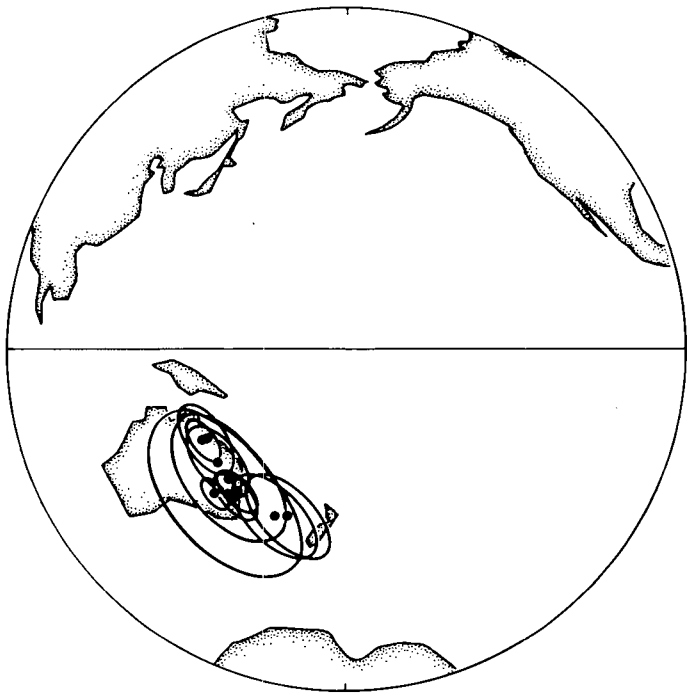


Figure 5. South virtual geomagnetic poles for reversely magnetized rock units from East Anacapa Island.

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TABLE 2. PALEOMAGNETIC DATA FROM EAST ANACAPA ISLAND

Unit	No.	Mean inclination	Mean declination	Kappa	Alpha 95	Virtual geomagnetic pole lat	Virtual geomagnetic pole long
Flow 10	4	−29.9	256.2	129.4	8.1	20.2 S	144.5 E
Flow 9	7	−40.2	244.5	266.2	3.7	33.1 S	143.7 E
Flow 8	8	−31.2	256.4	126.1	5.0	20.4 S	143.6 E
Flow 7	5	−32.9	242.0	200.4	5.4	32.8 S	150.3 E
Flow 6	5	−31.6	250.1	83.7	8.4	25.7 S	146.7 E
Flow 5	6	−34.6	241.4	126.8	6.0	33.9 S	149.5 E
Flow 4	3	−24.0	230.0	109.8	11.8	39.9 S	163.5 E
Flow 3	4	−27.9	232.2	63.2	11.6	39.4 S	159.5 E
Flow 2	3	−36.4	242.9	43.6	18.9	33.2 S	147.4 E
Flow 1	4	−32.1	245.5	38.4	15.0	29.7 S	148.9 E
Mean	10	−32.4	244.0	85.5	5.25	31.0 S	149.6 E
		32.4	64.0			31.0 N	30.4 W*

Note: data listed in order of decreasing age.
* Northern hemisphere equivalent.

TABLE 3. VIRTUAL GEOMAGNETIC POLES FOR SITES IN THE SANTA MONICA MOUNTAINS AND ANACAPA ISLAND

Site	Mean inclination	Mean declination	S‡	Kappa	Alpha 95	Virtual geomagnetic pole Lat	Virtual geomagnetic pole Long
ANACAPA	32.4	64.0	8.9	85.5	5.3	31.0 N	30.4 W*
PM	41.2	79.0	5.2	191.3	6.7	21.7 N	43.6 W
EC	41.3	81.0	10.8	75.2	10.7	20.1 N	44.4 W*
CP	35.8	59.2	—	47.1	4.6	36.1 N	29.6 W*
WPA	54.1	66.4	16.9	36.1	15.6	36.3 N	49.5 W*
WPB	57.0	54.4	21.6	25.4	13.5	46.4 N	49.6 W*†
REGIONAL VGP							
	41.3	69.8		54.1	10.5	29.2 N	39.6 W*
	42.0	68.0		40.5	12.2	30.8 N	39.4 W*†

Note: Virtual Geomagnetic Poles listed in order of decreasing age; Anacapa units may be younger.
* Northern Hemisphere equivalent.
† Flows 6 and 7 included.
‡ Angular standard deviation of directions from the mean site direction. WPA includes flow 6.

enough cooling units (time) have been sampled to average the secular variation. The angular standard deviation from the site mean for unit directions at each site is from 5° to 11°, whereas Cox's (1970) secular variation model predicts about 15° for this latitude (Table 3). Only at site WP is the dispersion high enough. However, when the angular standard deviation from the mean of the site directions is computed for all units from all sites, the dispersion is from 13° to 17°, depending on whether flows 6 and 7 at WP are included. This agreement of the unit dispersion data with the secular variation model may be due to two facts; first, that the set of individual units spans several million years in age and therefore average secular variation, and second, that the rotation may have occurred during the time span over which the units were formed, which would tend to increase the dispersion. The latter is probably the case at site WP. However, we cannot separate the two effects. Therefore, in averaging our directional data, we average both secular variation and rotation and translation.

Preliminary data from paleomagnetic measurements of rocks from Santa Cruz and San Miguel Islands and from the Tranquillon volcanics in the western Santa Ynez show between 65° to 130° of clockwise rotation. The concordance of these data with the Anacapa and Santa Monica results suggests that the entire northern Channel Islands ridge, including parts of the Santa Monica Mountains and possibly the Santa Ynez range, may have been rotated as a contiguous structural unit. The nearest northern boundary faults for this unit may be the Las Posas fault or the Oak Ridge fault (Fig. 1). Displacement along these faults is not well known and is complicated by Pliocene-Pleistocene compression (Crowell, 1976). Another northern boundary may be the Santa Ynez fault which Sylvester and Darrow (1979) have shown to be a major suture zone. The southern boundary faults are most likely the left-slip Malibu Coast and Santa Monica fault system. The eastern boundary may be at faults at the western end of the San Gabriel Mountains or beneath the San Fernando Valley–Verdugo Hills region; although samples from the Vasquez Formation also show clockwise rotations, suggesting that the boundary may be farther east, near the San Andreas. The western boundary fault system may lie to the west of San Miguel Island.

Freund (1974) and Dibblee (1977) suggested a tectonic model in which blocks between major right-slip faults rotate clockwise if bounded by conjugate left-slip faults. This model may be employed if left-lateral motion occurred on the east-west–trending faults during the rotation event. The timing of rotations suggests that they were almost completed in the late Miocene period. This timing corresponds fairly well with timing of displacement on the Malibu Coast fault. Truex (1976) states that Pliocene beds that overlap the

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Malibu Coast fault are evidence that movement on that fault had for the most part ceased by Mohnian time. In fact, the major motion seems to have occurred during a relatively short time. The Las Posas–Simi fault, which Truex (1976) considers to be the northern boundary of the Santa Monica terrane, also had its major period of movement before the late Miocene, with minor displacement during the Pliocene.

The paleomagnetic data suggest that a structural unit, or microplate, including the northern Channel Islands and the Santa Monica Mountains, rotated clockwise about a pivot point on its eastern end partly while being translated northward (Fig. 6). The data also generally agree with a hypothesis of Crouch (1979, this volume) who suggests that the western Transverse Ranges and the outer borderland off southern California originated from the southern borderland off Baja California. Krause (1965) previously noted a deep-water region off northern Baja California, and Suppe (1970) suggested that it was once filled by portions of the Salinain block and some Transverse Ranges blocks, including an analogue of the structural unit discussed here. Although Suppe suggested rifting from the region, he did not note rotation. He believed that the rifting there was almost complete in the late Oligocene or early Miocene. We cannot be certain yet whether the Santa Ynez range was part of this rotated block. Our existing paleomagnetic data do support post-Miocene rotation here, but the remanent inclination results do not require northward transport.

We have considered three possible prerotation positions for the rotated block relative to the continental margin (Fig. 6). Notice that about 120° of total rotation are indicated for all of the possibilities, while we generally observed less than this. This could mean that more rotation did in fact occur and that it began before middle Miocene time. This possibility awaits testing in older rocks.

For position A, the block has been swung back along the coast without southward movement. This is contrary to our mean paleolatitude estimate of 24° ± 9° (s.d.) degrees, which would call for about 10° of latitude change, but within the accuracy of the estimate. However, the position is more consistent with the findings of Howell and others (1974) and Abbott and Smith (1978), who have correlated Eocene gravels on Santa Cruz Island with the Poway conglomerate at San Diego. Position B is that of Crouch (1979, this volume) where various pieces of the borderland have been placed off northern Baja California, with the western Transverse Ranges being the most northerly. Position B indicates about 5° of northward movement for the Santa Monica Mountains region, which agrees generally with our data but is less than the average paleolatitude estimate. In position C, the block is moved into the void of Vizcaino Bay. This is probably the most southerly position it could occupy and would require about 6.5° of northward movement. However, southward restoration of 1° or 2° movement on a proposed Transpeninsular Baja fault (Crouch, 1979, this volume) would bring the paleolatitudes for the Santa Monica region to about 26°N, indicating more than 8° of latitude change. Our paleomagnetic data best support position C, assuming no more than 4° northward movement of Baja California and southern California since middle Miocene time. If northward movement occurred prior to the opening of the Gulf of California, relative movement between the continental margin and the rotated area would be less. In any case, within the limits of the paleolatitude estimate, the rotated rocks could have originated anywhere in the borderland.

Positions A, B, and C (Fig. 6) all fell between the Mendocino and Rivera triple junctions (in the San Andreas system) 15 m.y. ago (Dickinson and Snyder, 1979; Atwater, 1970; Atwater and Molnar, 1973). Therefore, rotation and translation of the block was the result of Pacific–North American plate interaction. If pre-middle Miocene rotations did occur, then they may have resulted from Farallon–North American plate interactions.

The amount of rotation observed implies that crustal or plate convergence has occurred on the leading edge of the rotated block, and rifting occurred near its trailing edge as it swung into its present position. This rifting may have formed a rhombochasm such as

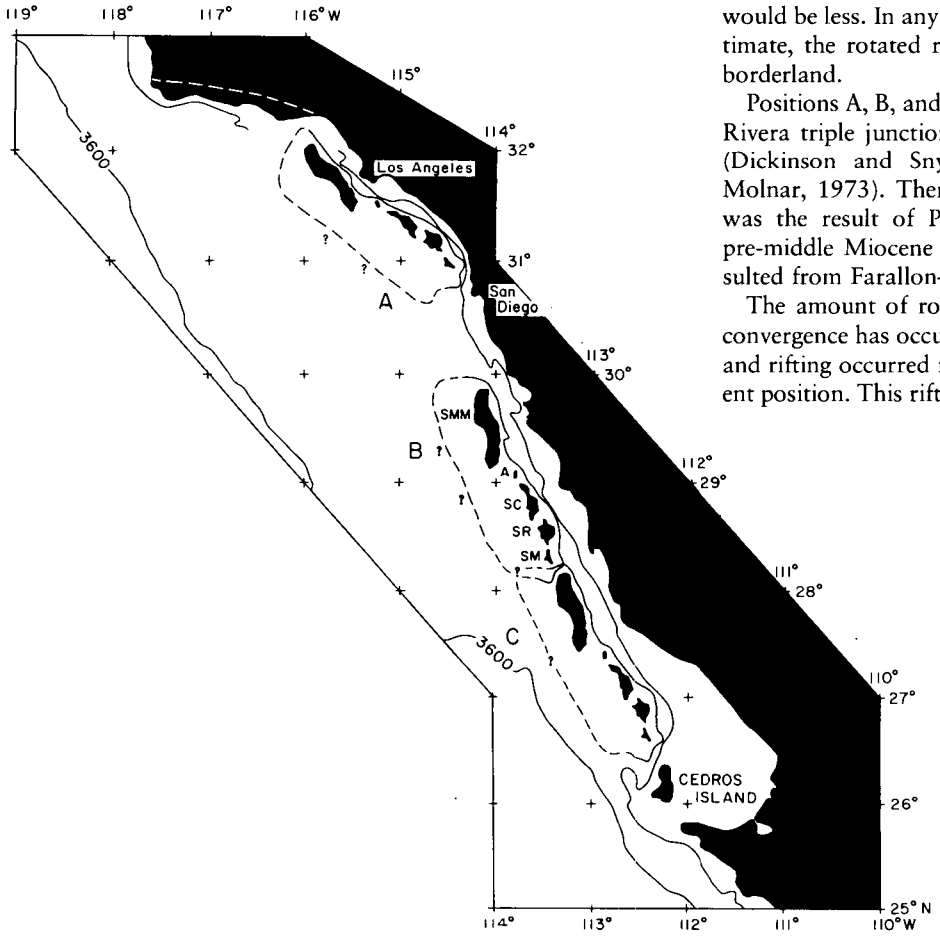


Figure 6. Three possible origin sites for the rotated block (A, B, C). Base map shows the Gulf of California closed relative to a fixed North America. The paleogeography of the other channel islands is not considered in this discussion.

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the Los Angeles and other deep sedimentary basins. With the exception of Eocene strata on southwest Santa Cruz Island, restoring the rotated block back along the continental edge brings the generally northward-trending paleocurrent directions in Eocene and Oligocene rocks within the block (Sespe, Poway conglomerate, and equivalents) to a westward direction. This is consistent with surrounding data, and it eliminates the need to reconstruct the Poway submarine fan, as proposed by Yeats and others (1974).

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